EDM cutting plans

User guide of the cutting-plan tool (`cutting-sheets/`) – version of 03 November 2026

1. What the tool does

You write a short order (a text file: which specimens, how many, at which angle to the rolling direction, on which sheet). The tool then, without any manual step:

Step	What happens
1. Order and templates	reads and checks the order; takes the template of each specimen type from the lab catalogue
2. Specimens	turns each specimen to its angle, adds its notch code (angle + index) and its 2 Mikrosteg recesses
3. Placement	places all the specimens on as few sheets as possible: 5 mm gap, inside the effective area of the machine, never mirrored
4. DXF	writes the cutting plan (mm, colours = EDM quality Q3 / Q4 / Q5)
5. Checks	reads the final DXF back and checks contours, gaps, margins, outlines and codes
6. Notice and registry	writes the notice for the workshop and the specimen registry

Typical durations: example order (14 specimens) about 15 s; real CT sheet (68 specimens) about 1.5 min.

2. Installation (once per PC)

The tool runs on Windows with Python 3.10 to 3.14 (64 bit) and Git Bash. Open Git Bash and check the Python version:

```
py --version
```

Then install the Python modules:

```
py -m pip install ezdxf shapely matplotlib scipy pyyaml spyrrow==0.9.0 packingsolver
```

Get the repository the first time (then `git pull` is enough to update it):

```
cd ~/Documents
git clone https://github.com/josselindavion/eth.git
```

> If py starts an old Python (for example 3.7, 32 bit), spyrrow cannot be installed: install a recent Python 3 (64 bit) from python.org and run `py -0p` to list the installed versions.

3. Running an order

Always from the repository root:

```
cd ~/Documents/eth
git pull
py cutting-sheets/scripts/make_plan.py cutting-sheets/orders/example.yaml
```

Option	Effect
(none)	results in `cutting-sheets/runs/<name>/`; this folder is ignored by git, so runs can be repeated freely
`--final`	results in `cutting-sheets/output/<name>/` (kept by git): use it for the plan sent to the workshop, then `git add`, `git commit`, `git push`
`--time 5`	replaces `time` of the order (seconds of compaction per sheet)

The terminal shows the 6 steps, progress bars with the remaining time, the specimens moved between sheets, and the table of checks. A good run ends with:

```
DONE, all checks OK, in 13 s
    Plan    : .../runs/ETH_400x300mm_SHEET_v20261103_example/ETH_400x300mm_..._example.dxf
    Notice  : .../notice.pdf
    Registry: .../registry.csv
    Preview : .../preview.png
```

4. Writing an order

Copy `cutting-sheets/orders/example.yaml` under a new name and edit it with any text editor. Every field is explained by a comment in the file. Text after # is a comment. Keep the indentation (spaces, no tabs).

```
name: ETH_400x300mm_SHEET_v20261103_UT30   # plan name: output folder, DXF, notice
sheet: 400x300                      # complete sheet, width x height (height = rolling direction)
clamping: {left: 40, bottom: 40}    # L-shaped clamping
machine_area: 350x250               # wire travel of the EDM (AgieCharmilles CUT E 350)
placement_margin: 5                 # tolerance on the position of the sheet in the machine
edge_margin: 5                      # distance to a free edge of the sheet
gap: 5                              # distance between two specimens
specimens:
  - {type: UTUT, angle: 30, count: 5}
  - {type: NT20, angle: 45, count: 3, first_index: 1}
```

4.1 The sheet on the machine

Field	Meaning	Default
sheet	complete sheet, width × height in mm; the height is the rolling direction	required
clamping	bands held by the clamps, per side (left, bottom, right, top); {left: 40, bottom: 40} = L-shaped clamping	none
machine_area	area the wire can reach (CUT E 350: 350x250 = X × Y travel)	required
machine_position	where the machine area starts on the sheet, {left: .., bottom: ..}	inner corner of the clamping
placement_margin	nothing is placed within this distance of the machine area edge, because the sheet is not positioned to the mm	5
edge_margin	minimum distance to a free (not clamped) edge of the sheet	5
gap	minimum distance between two specimens	5

Effective area (where the specimens go) = machine area minus the placement margin on each side, inside the sheet and outside the clamping. With the values above: machine area from (40, 40) to (390, 290), effective area 340 × 240 mm from (45, 45). The terminal, the notice and the preview show these zones.

4.2 Placement time

Field	Meaning	Default
assign_time	seconds for PackingSolver to put all the specimens on sheets	30
fill_tries	failed tries allowed when moving specimens to empty one sheet (0 = faster, maybe one sheet more)	8
fill_time	seconds at most per try (a try that succeeds stops after 1–3 s)	10
time	seconds at most of final compaction per sheet (0 = none)	15
seed	random seed: change it to get another layout	1
max_sheets	the order is rejected if it needs more sheets	20

4.3 Specimens

One line = count specimens of one type at one angle.

Key	Meaning
type	specimen type (table below)
angle	direction of the coded end from the rolling direction, counter-clockwise: 0 to 135° in steps of 15° (written in the notch code)
count	number of specimens
first_index	index (number in the notch code) of the first specimen, then +1 each (default 0, max 9)

Type	Short name	Specimen
UTUT	UT	uniaxial tension, 106 mm
NT20 / NT06	NT6.67 (for NT06)	notched tension, radius 20 / 6.67 mm, 106 mm
CHCH	CH	central hole, 106 mm
UTST	–	short uniaxial tension, 81 mm
SHLD / SHMD / SHHD	–	short shear, 81 mm
MNAK_FULL / MNAK_HOLE	DISC0 / DISC2	Mini-Nakajima disc without / with 2 holes (holes at angle + 90°)
MNAK_RECT_FULL / MNAK_RECT_HOLE	– / MNAK_RECT	rectangular Mini-Nakajima 80 × 54 mm without / with 2 holes (holes oriented as on the disc with the same angle)
TENS_FULL	STRIP	strip 140 × 30 mm, no code

Only the fields of this guide are accepted: an unknown field (typo, old field) stops the run with the list of allowed fields, so nothing is silently ignored.

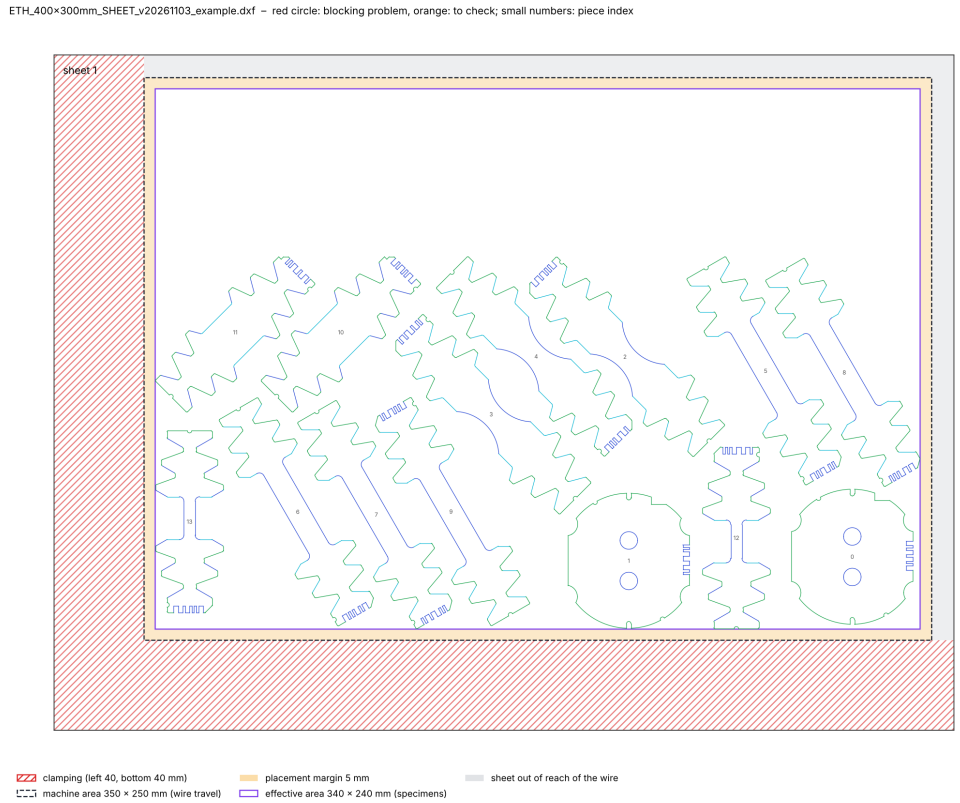
5. Results

Each run writes exactly 4 files in its folder:

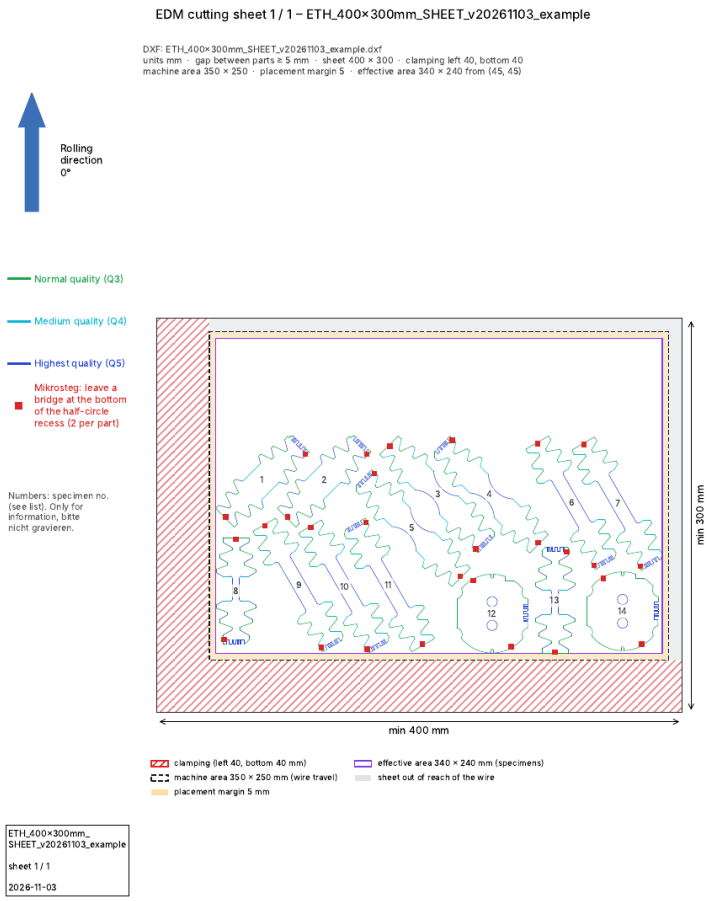
File	For	Content
<name>.dxf	workshop	the cutting plan: one frame per sheet, every specimen with its notch code; mm; colour = EDM quality (3 = Q3 normal, 4 = Q4 medium, 5 = Q5 high)
notice.pdf	workshop	one page per sheet (numbered specimens, rolling direction, qualities, zones, Mikrostegs as red squares) and the list of specimens
registry.csv	lab	one line per specimen: number, sheet, type, angle, notch code, position, specimen ID
preview.png	you	all the sheets with the zones, to check the plan at a glance

Mikrostegs: every specimen has 2 small half-circle recesses (radius 1 mm) in its outline. The workshop stops the wire at the bottom of each recess (red square on the notice), so the specimen stays attached; after breaking it out, the rest of the bridge stays inside the recess.

Send the DXF and notice.pdf to the workshop; keep registry.csv for the lab.



preview.png of the example order. Red hatching: clamping; black dashed: machine area (wire travel); orange band: placement margin; purple: effective area where the specimens are placed; grey: part of the sheet out of reach of the wire. Line colours = EDM quality.



First page of notice.pdf (example order).

6. Checks and error messages

After writing the DXF, the tool reads it back without any assumption. Every line must be OK:

Check	What is verified
Units	the DXF is in millimetres
Sheets	one frame per sheet, of the ordered size
Closed contours / duplicate lines	every cutting contour is closed, no line twice
Colours = EDM quality	cutting lines only in colours 3 / 4 / 5
Overlaps / gap between pieces	no overlap; at least gap mm between two pieces
Margin to sheet edge	every piece inside the effective area
ID codes	every coded specimen carries a notch code; shared codes only between different shapes
Specimens vs order	outline = template, code read back = code ordered, direction = angle ordered

Message	Meaning and what to do
ORDER REJECTED: unknown field(s): …	typo or old field in the order: use the names of section 4
ORDER REJECTED: missing field "machine_area"	sheet and machine_area are required
ORDER REJECTED: no effective area left	clamping + margins leave no room: check sheet, clamping, machine_area, placement_margin
ORDER REJECTED: … does not fit in the usable area	a specimen is larger than the effective area once turned: bigger sheet or smaller margins
ORDER REJECTED: the specimens need N sheets	more sheets than max_sheets: raise max_sheets or split the order
DONE WITH ERRORS (see above)	a check failed: do not send the plan; the preview circles the problem in red
py314: command not found	use py (the alias py314 only existed in one terminal)

7. Tips

I want…	Do this
a faster run	lower fill_tries (0 = no try to remove a sheet) and time
fewer sheets	raise fill_tries / fill_time, or try another seed
a different sheet or clamping	change sheet, clamping, placement_margin: the preview shows the new zones
to keep the plan sent to the workshop	run again with --final, then git add / commit / push

More details (method, engines, notch code): `cutting-sheets/README.md`.